

Lab: Building a POMDP

Learning Goal: Construct and analyze a simple POMDP from scratch.

Part 1: Constructing the A Matrix

Scenario: A foraging agent in a 2-location world. States: {food_left, food_right}.

Observations: {see_food, see_nothing}.

1. Design an A matrix where the agent can partially observe the food location (occasionally gets it wrong).
2. Verify each column sums to 1.
3. What would the A matrix look like if observation were perfect? Noisy?

	food_left	food_right
see_food		
see_nothing		

Write your response here

Part 2: Constructing the B Matrix

Learning Goal: Build transition dynamics with agency.

Same scenario: Actions: {go_left, go_right, stay}.

Design the B matrix for each action. The agent is currently at the right location.

For action "go_left":

	food_left	food_right
food_left		
food_right		

Do the same for "go_right" and "stay."

Write your response here

Part 3: Setting Preferences (C Vector)

Learning Goal: Translate goals into the C vector.

Exercise: Define C vectors for different types of agents:

1. A hungry agent that wants food: $C = ?$
2. A scared agent that avoids everything: $C = ?$
3. A curious agent that wants novelty (any observation is fine): $C = ?$

Write your response here

Part 4: Running the Perception-Action Loop

Learning Goal: Manually trace one cycle of belief updating and action selection.

Setup:

- $D = [0.5, 0.5]$ (no initial preference for food location)

- A, B, C as you defined above
- Agent observes: see_food
- Update beliefs: $q(s) \propto A(\text{see_food}, :) \times D$
- Normalize $q(s)$ so it sums to 1
- Based on updated beliefs, which action should the agent take?
- How would your answer change if the observation were "see_nothing"?

Write your response here

Part 5: Reflection

In 150 words, reflect: The POMDP framework uses just 5 components (A, B, C, D, π) to describe an agent's entire decision-making system. What is gained by this simplicity? What might be lost?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Matrix construction	A matrix (observation model)
2	Matrix construction	B matrix (transition model)
3	Preference encoding	C vector (preferences/values)
4	Loop tracing	Perception-action cycle
5	Framework reflection	Strengths and limits of POMDPs